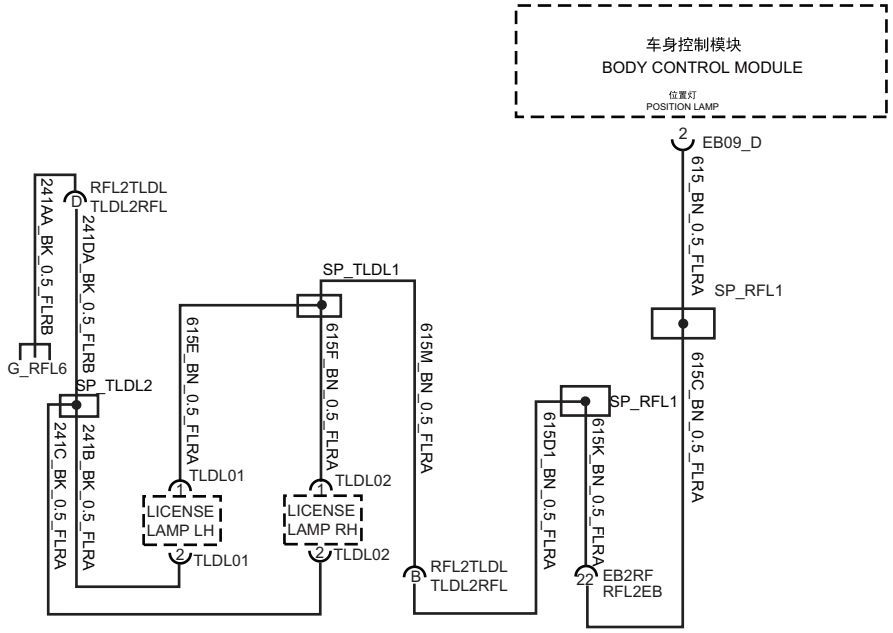
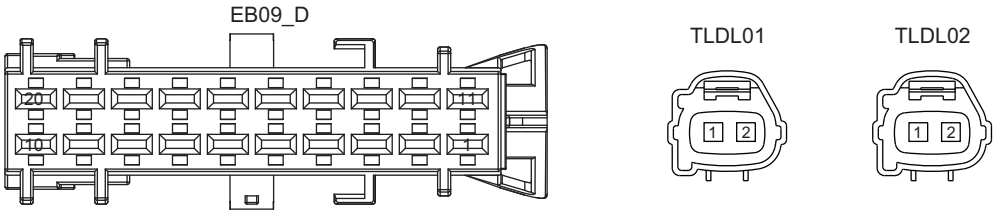
B1220-11、 B1220-12、 B1220-13

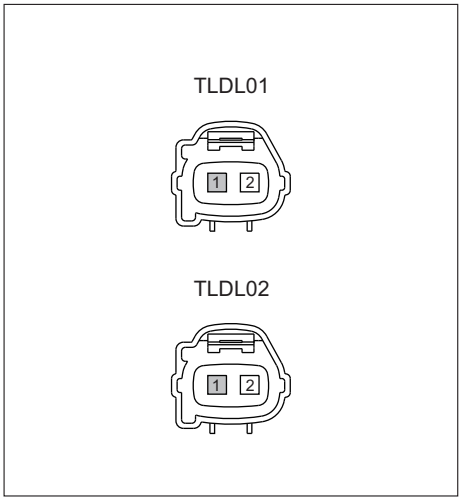
Fault diagnosis code
B1220-11：License plate light circuit short to ground
B1220-12：License plate light circuit short to battery
B1220-13：License plate light circuit open
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • BCM fault • License lamp fault • Insurance fuse fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
B1220-11
License lamp output circuit short to ground
B1220-12
License lamp output circuit short to battery
B1220-13
License lamp output open
To set the effect of a fault code condition
License lamp function fails to effect
Description of circuit diagram

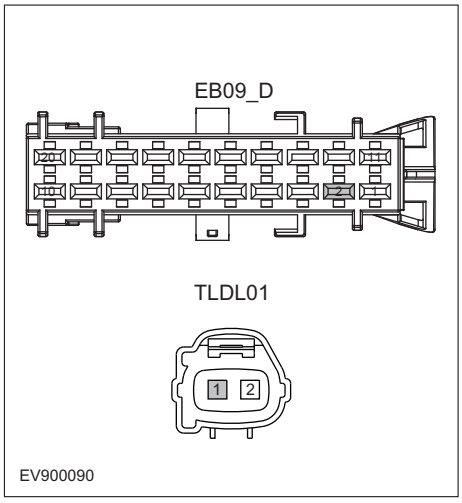
Circuit diagram

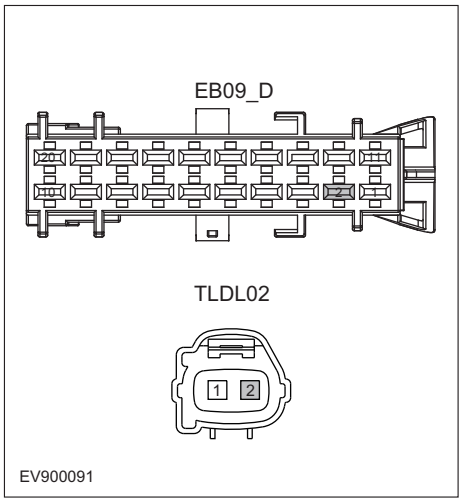


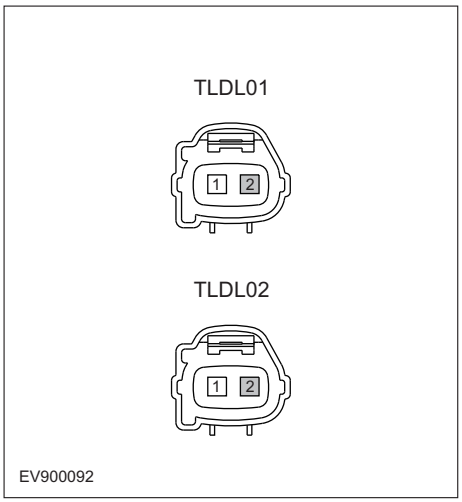


EV900088

Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of air conditioning control module for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Check the body control module with a diagnostic apparatus. 4 Is there any other fault code except for B1220-11/B1220-12/B1220-13?
		YES → Refer to:DTC Summary list(BCM).
NO ↓		
Step 3	Check the supply power of license lamp when it is started or operating.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug TLDL01 of left front license lamp and TLDL02 of right front license lamp. 3 Turn the ignition switch to ON. 4 Measure the voltage between the harness terminal of license lamp and ground. Measuring circuit: voltage between terminal 1 on plug TLDL01 and ground. Measuring circuit: voltage between terminal 1 on plug TLDL02 and ground. **Standard value:10~14V** 5 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		

Step 4	Check the circuit between left front license lamp and BCM (open).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB09_D of BCM. 3 Measure the resistance between the harness terminal of left front license lamp and of BCM. Measuring circuit: resistance between terminal 1 on plug TLDL01 and terminal 2 on plug EB09_D. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 5	Check the circuit between right front license lamp and BCM (open).
	1 Measure the resistance between the harness terminal of right front license lamp and of BCM. Measuring circuit: resistance between terminal 1 on plug TLDL02 and terminal 2 on plug EB09_D. **Standard value: ≈ 0Ω** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Test whether the grounding circuit of license lamp is open.
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness terminal of license lamp and ground. Measuring circuit: resistance between terminal 2 on plug TLDL01 and ground. Measuring circuit: resistance between terminal 2 on plug TLDL02 and ground. **Standard value: < 1Ω** 3 Test whether the resistance is less than 1Ω?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 7	Test whether BCM is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB09_C、EB09_D、IP09_A、IP09_B of BCM. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BCM plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the BCM.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	
Next Step ↓	
Diagnosis end.	